FREE RESOURCE I THE MBA PA

Cardiac Surgery First-Assist Protocol Template

A Practical Readiness Framework for Cardiac Surgery PAs, Surgical Techs, and Surgical Residents

By Pol Martin D. Vedar, PA-C, MBA

The MBA PA

A practical readiness framework for cardiac surgery first assistants — so new APPs earn trust faster and programs scale training beyond any one person. Built from 13+ years of OR experience and refined across multiple cardiac surgery programs.

How to Use This Resource

- **For individual clinicians:** Use the five pre-case questions and intraoperative standards as a daily readiness checklist. The debrief template turns every case into a learning event.
- **For preceptors:** Use the common failure points to diagnose where a trainee is stuck — and the competency checklist to document progression objectively.
- **For program leaders:** Use the program-level competency checklist to standardize what "ready" means across your entire APP team — no more tribal knowledge.

---

1) First-Assist Readiness: Pre-Case Protocol

Before entering the room, the first assistant should be able to answer five questions clearly.

Question 1

What procedure are we doing — and what's the plan?

Know the specific operation (CABG x3? AVR with root enlargement? Redo sternotomy?). Understand the surgeon's preferred sequence: cannulation strategy, conduit plan, valve sizing approach. A first assist who knows the plan can anticipate the next move — and that's what builds trust.

Question 2

What's unique about this patient?

Review the chart for anatomy variants (redundant colon for conduit? porcelain aorta?), comorbidities that change the game (EF <20%? pulmonary hypertension? prior chest radiation?), and anything that makes this case different from a textbook version. The surgeon should never be surprised by something you could have flagged.

Question 3

What equipment is needed — and is it in the room?

Check: sternotomy saw tested? EVH tower positioned and functional? Valve sizers and implant kit confirmed with rep? Cell saver set up? Defibrillator pads placed? Hemostatic agents available? Assume nothing. Verify everything.

Question 4

What are my specific responsibilities in this case?

Will you be harvesting conduit? First-assisting the entire case? Managing hemostasis during closure? Know your role before the drapes go up. If you're harvesting and first-assisting, clarify the handoff point with your backup. Unclear roles create OR delays — and delays erode confidence.

Question 5

What's the most likely complication — and what's my role in managing it?

For every case type, identify the most common intraoperative emergency: CABG → graft spasm or poor flow; AVR → annular disruption; Redo → cardiac injury on re-entry. Know what you'll do and what you'll hand the surgeon before it happens. Preparation is the difference between panic and performance.

---

2) Intraoperative Standards

Four pillars of effective first-assisting — measurable, teachable, and consistent across every case.

PILLAR	STANDARD	WHAT "GOOD" LOOKS LIKE
Exposure	Provide and maintain optimal visualization of the operative field at all times.	Surgeon never asks for adjustment twice. Suction is proactive, not reactive. Field is dry before the surgeon looks up.
Anticipation	Know what instrument, suture, or move comes next — before it's requested.	Instruments in hand before the surgeon extends a palm. Suture loaded before the last throw is cut. No verbal requests needed for routine steps.
Hemostasis	Identify and address bleeding points independently within your scope.	Bleeding managed at the assistant level whenever appropriate. Surgeon notified of anything beyond your scope — with a specific description, not a vague "it's bleeding."
Communication	Closed-loop communication for every critical step and transition.	"Coming off pump" acknowledged. "Protamine in" confirmed. Conduit quality concerns raised immediately — not after the anastomosis is done.

3) Five Most Common First-Assist Failure Points

And fast corrections that turn them into growth opportunities.

1. **Passive suctioning — waiting for the field to flood before acting**

   Correction: Develop a rhythm: suction on every instrument exchange. Keep the field dry as a habit, not a reaction. If the surgeon has to ask for suction, you're already behind.

2. **Instrument paralysis — freezing when the surgeon asks for something unfamiliar**

   Correction: Study the instrument tray before every case. If you don't know a name, learn it after the case. Keep a personal instrument guide. Never say "I don't know what that is" twice for the same instrument.

3. **Suture management gaps — losing tension, tangling, or loading the wrong suture**

   Correction: Maintain gentle tension on every suture — not enough to tear, enough to keep the field organized. Confirm suture type and needle with circulator before loading. Organize suture ends in a consistent pattern.

4. **Silence during complications — going quiet exactly when communication matters most**

   Correction: In a crisis, your voice is a tool. Call out what you see: "Pressure's dropping — looks like the graft." Closed-loop every instruction. Silence in an emergency reads as panic — even when it's just concentration.

5. **No debrief — finishing the case and moving on without extracting the lesson**

   Correction: Use the post-case debrief template (Section 4). One question per case: "What will I do differently next time?" That habit — over 100 cases — is what separates a competent first assist from an exceptional one.

---

4) Post-Case Debrief Template

Turn every case into deliberate practice. Complete within 24 hours while memory is fresh.

Structured Debrief Prompts

1. **What went well today?**
   Be specific. Not "the case went fine." Identify one moment where your preparation or skill made a difference.

2. **What would I do differently?**
   Identify one specific moment. Was it instrument knowledge? Exposure technique? Communication timing? The more precise, the more actionable.

3. **What did the surgeon do that I want to incorporate?**
   Watch your surgeon's decision-making, not just their hands. When did they change the plan? Why? What judgment call did they make — and what would you have done?

4. **Was there a moment I wasn't ready?**
   Honesty here is the fastest path to growth. A moment of being unprepared isn't failure — it's data. Log it. Address it before the next similar case.

5. **What's my one focus for the next case?**
   Pick one thing. Not three. Deliberate practice means improving one specific skill at a time. Write it down. Review it before the next case.

CASE

Procedure: ____________________

Date: ____________________

SELF-RATING

Exposure: 1 2 3 4 5

Anticipation: 1 2 3 4 5

Communication: 1 2 3 4 5

5) Program-Level Competency Checklist

Standardize onboarding and training across your entire APP team. Check off each item when the APP demonstrates consistent, independent competence — not just one successful case.

Phase 1: Foundation (First 30 Days)

- [ ] **OR orientation complete:** Knows room layout, equipment locations, emergency protocols, and circulator/perfusion team introductions.
- [ ] **Instrument recognition:** Can identify and retrieve every instrument on the cardiac tray by name — without prompting.
- [ ] **Sternal closure:** Independently closes sternum with wire under supervision. Demonstrates appropriate tension and wire spacing.
- [ ] **Chest tube placement:** Independently places and secures chest tubes. Confirms placement with surgeon.

Phase 2: First-Assist Integration (Days 31–60)

- [ ] **Cannulation assist:** Assists with aortic and venous cannulation — purse-strings, snares, cannula insertion, de-airing. Understands sequence for standard and minimally invasive approaches.
- [ ] **Proximal anastomosis assist:** Provides exposure for proximal vein graft anastomoses. Anticipates partial-occlusion clamp placement and removal.
- [ ] **Valve case assist:** Assists with valve excision, annular debridement, sizing, and implant. Understands differences between mechanical, bioprosthetic, and TAVR cases.
- [ ] **Coming off bypass:** Assists with de-airing, pacing wire placement, and hemodynamic transition. Communicates rhythm, bleeding, and volume status proactively.

Phase 3: Independent First-Assist (Days 61–90)

- [ ] **Independent CABG first-assist:** Handles entire CABG case as primary first assist — cannulation through closure — without preceptor prompting.
- [ ] **Independent valve first-assist:** Handles AVR and/or MVR as primary first assist across at least 10 cases.
- [ ] **Re-entry / emergency readiness:** Understands redo sternotomy protocol, emergency cannulation sites, and internal cardiac massage access. Can function as second assist in emergency re-entry.
- [ ] **Conduit harvesting competency:** If harvesting is part of role: independently harvests vein and/or radial conduit with ≥90% Grade A or B outcomes.

Sign-Off

This checklist is complete when all items are checked, dated, and signed by both the trainee and the Lead APP or designated preceptor. Sign-off represents readiness for independent first-assist duties in the specified case types — not the end of learning.

TRAINEE

Name: ____________________
Signature: ____________________
Date: ____________________

LEAD APP / PRECEPTOR

Name: ____________________
Signature: ____________________
Date: ____________________

The MBA PA

Pol Vedar, PA-C, MBA — Cardiothoracic Surgery

pol@themba-pa.com | themba-pa.com